

Test 1: b(N) convergence table

Verdict: PASS | Generated: 2026-09-02 13:48:07 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

b(N) = (1/N)?|?_k???_k| measures convergence of Gram-point approximations to true zeta zeros. EXACT-formula family: tolerance scale = floating-point round-off.

METHOD: b(N) = (1/N)?|?_k???_k| measures convergence of Gram-point approximations to true zeta zeros. EXACT-formula family: tolerance scale = floating-point round-off.

Result

Test 1 [HARDCORE pass 2]: 8 sub-checks, 0 failed, b(50000)=1.2126 ? PASS

'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/plot_02' (.png 600dpi / .svg / .pdf)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 1 --no-two-pass

Independent verification guide

Deep computation logs in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(no log_comp entries)

FULL COMPUTATION LOG

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CONSOLE CAPTURE

TEST 1 (secondary, HARDCORE): Objection 1: b(N) Convergence ? deep validation

Same statistic b(N) = (1/N) ?|?_k ? ??_k|, verified by 8 harder sub-checks

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H0 reproduce : max|b_direct ? b_prefix| over 4 probes = 0.00e+00 (limit 1e-9)

H1 dense scan: 40 checkpoints, N = 25 ? 50000 (ratio 1.22)

N	b(N)	? vs prev
25	4.6101190361	---
30	4.3989811028	-0.2111379332
37	4.1932852303	-0.2056958726
45	3.9943669591	-0.1989182711
55	3.8252166187	-0.1691503404
67	3.6455743228	-0.1796422960
82	3.4801726407	-0.1654016820
100	3.3321638710	-0.1480087697
122	3.1910285619	-0.1411353091

149	3.0547427984	-0.1362857635
182	2.9267044466	-0.1280383518
222	2.8066104527	-0.1200939939
271	2.6947048268	-0.1119056258
331	2.5892260492	-0.1054787776
404	2.4907320942	-0.0984939550
493	2.3966440338	-0.0940880604
601	2.3100937127	-0.0865503211
733	2.2274677461	-0.0826259666
894	2.1498825733	-0.0775851728
1091	2.0754888256	-0.0743937476
1331	2.0067062433	-0.0687825824
1624	1.9405012929	-0.0662049503
1981	1.8782421944	-0.0622590986
2417	1.8197691194	-0.0584730750
2949	1.7640375461	-0.0557315733
3598	1.7111812347	-0.0528563114
4390	1.6613158059	-0.0498654288
5356	1.6137261550	-0.0475896509
6534	1.5686043745	-0.0451217805
7971	1.5258165855	-0.0427877890
9725	1.4850058741	-0.0408107114
11864	1.4460749475	-0.0389309266
14474	1.4090310622	-0.0370438853
17658	1.3736819777	-0.0353490845
21543	1.3399113218	-0.0337706559
26282	1.3076687539	-0.0322425680
32064	1.2768213946	-0.0308473593
39118	1.2473022976	-0.0295190970
47724	1.2190083740	-0.0282939236
50000	1.2125526773	-0.0064556968

H2 monotone : 0/39 consecutive pairs exceed the noise band (limit 2)

H3 predict : walk-forward one-step-ahead, best of 3 laws (power / 1•log?,„N / local secant)

median err 0.05%, max err 0.09% (limits 2%/5%)

H4 stationarity: head b(25000)=1.315628 | tail b(25000)=1.109478 | tail/head = 0.8433 (limit 1.25)

H5 bootstrap : b(50000)=1.212553 | 95%CI [1.209765, 1.215265] (width 0.45% of b, limit 15%)

threshold 2.00 ? (b + 2?/?N = 1.215350) = +0.784650 ? verdict ROBUST

H6 outliers : peak |??| = 9.0356 at k=1 (0.0% of ?d, limit 10%)

top-5 share 0.1% | 1%-trimmed b deviates +1.13% from b (limit 15%)

H7 Gram sanity: ?? strictly increasing over n=1..50000 ? 0 violations (limit 0)

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H0 reproduce PASS ? worst |?| = 0.00e+00

H1 dense scan PASS ? 40 checkpoints

H2 monotone PASS ? 0 violations / 39 pairs

H3 out-of-sample PASS ? walk-forward median 0.05%, max 0.09%

H4 stationarity PASS ? tail/head = 0.8433

H5 bootstrap PASS ? CI width 0.45%, margin +0.7846

H6 outlier audit PASS ? peak 0.0% of ?d, trimmed dev 1.13%

H7 Gram reference PASS ? 0 violations

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